

ESP Well Troubleshooting Cases



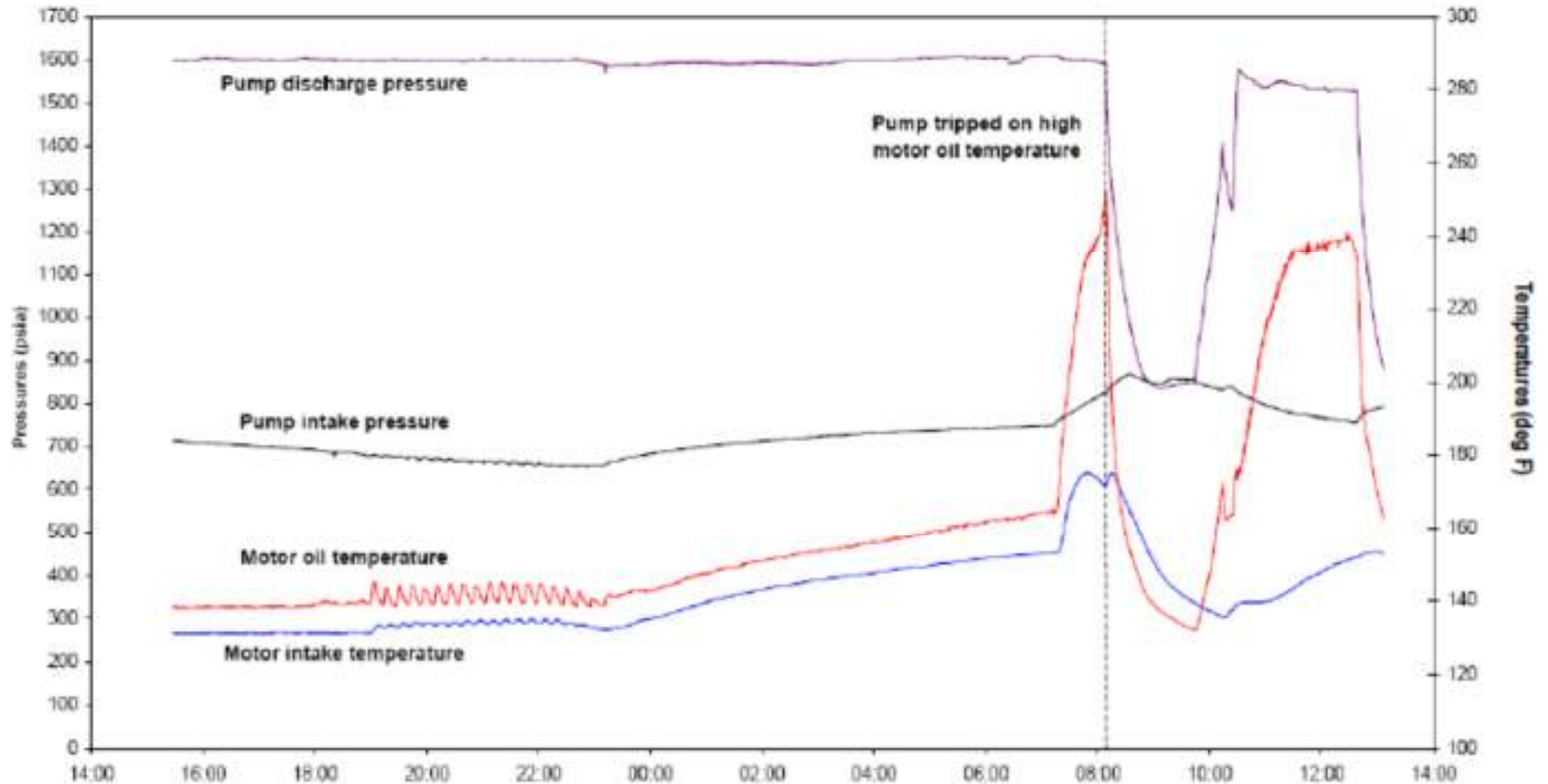
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Objectives of this Module

Upon completion of this section, the participant will be able to:

- Analyze operating data and determine deviations from the normal operation.
- Generate recommendations to fix operating problems in Electric Submersible Pump Systems.
- Suggest actions to optimize the operation of a ESP System.

Pressure and Temperature Trends



Summary of Findings

Gas Locking in ESP motor shroud

Summary of Findings

- ESP has 7" shroud (6.18" ID) attached with gravel pack screens below motor (5.62" OD).
- Pumping running normally until 23:00 hrs when pump ΔP starts to decrease slowly.
 - No change in pump discharge pressure indicates no change of fluid density in tubing.
 - Confirmed no change to motor speed.
 - Slow decrease in pump ΔP points to increasing quantities of free gas in pump.
 - Pump continues to process gas until 08:00 hrs when the high motor temperature shut down protects the ESP
- After pump shut down, fluid level in tubing falls hence drop in discharge pressure.
- Pump restarted at 10:00 hrs but same rapid rise in motor temperature

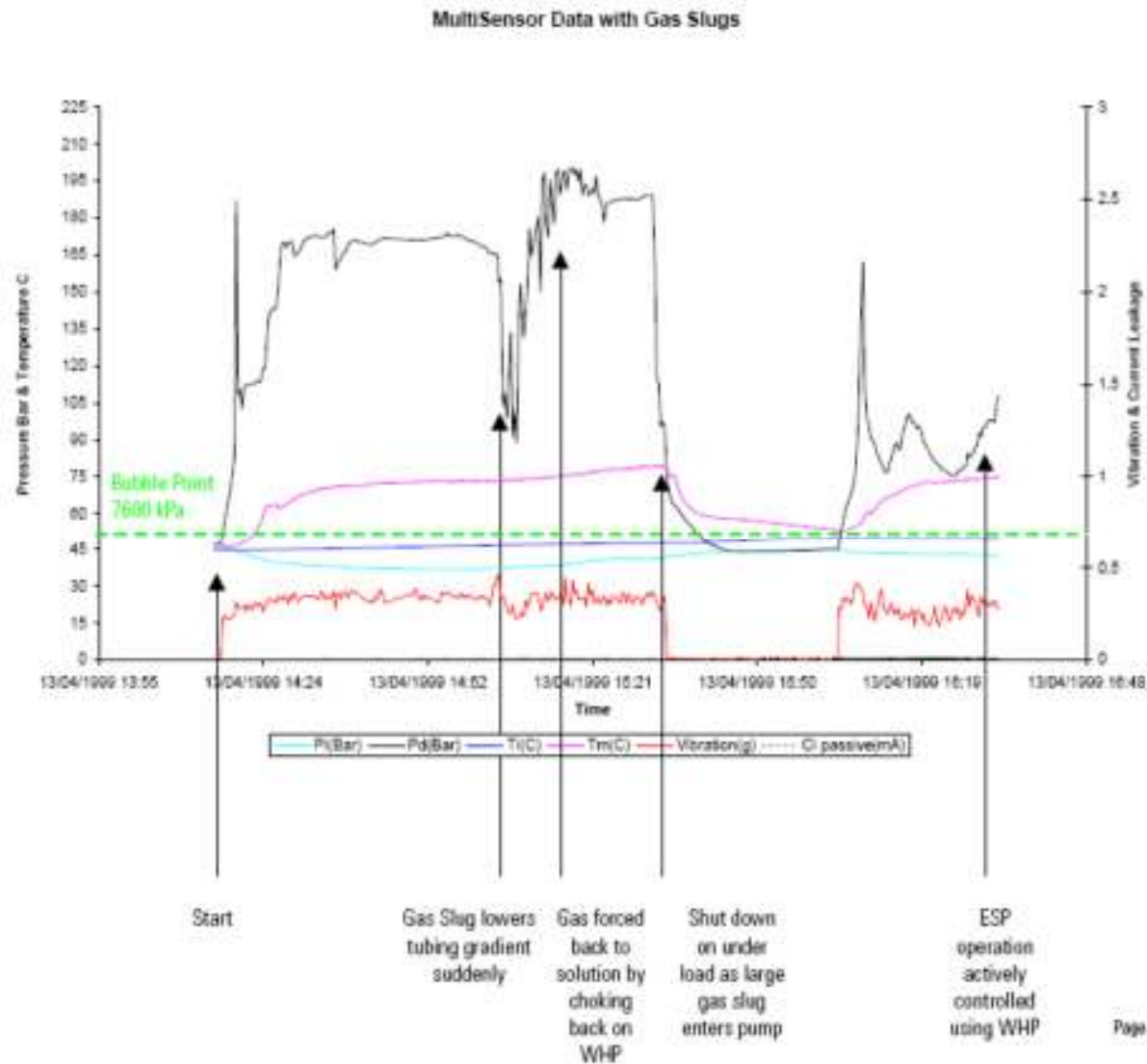
Diagnosis

- Loss of pump ΔP indicates presence of free gas.
- Fluid bubble point is approximately 30 psia below the lowest intake pressure observed. However inside the motor shroud with heavy oil, low heat transfer and elevated skin temperatures bubble point increases with temperature and so free gas emerges inside the shroud.
- Pump can handle this small amount of gas without gas locking.
- But small annulus (0.25") between ESP motor and shroud fills with gas, stopping heat transfer away from motor.

Using the MultiSensor data on high GOR wells

The example trend below shows a standard ESP with gas separator running in a well with high free gas content. The well is horizontal and “burps” large slugs of gas, which upset the ESP operation. The only way to control these slugs was to quickly choke back the wellhead pressure as discharge pressure was seen to drop off (thereby compressing some of the discharge pressure of the pump. This caused the pump to run in severe up thrust both from the lack of back pressure and also the low gravity of fluid being pumped.

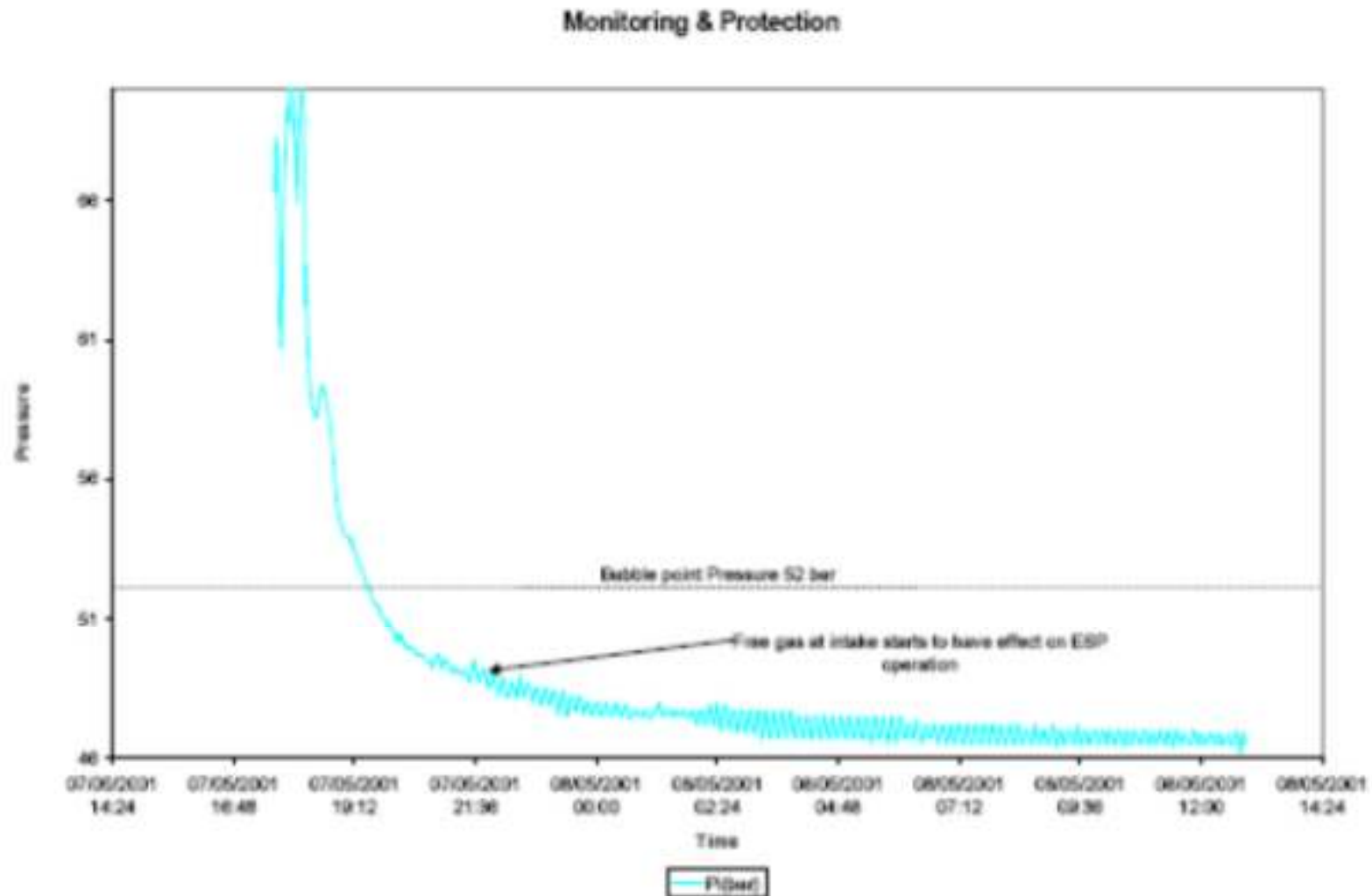
Using the MultiSensor data on high GOR wells



Using the MultiSensor data on high GOR wells

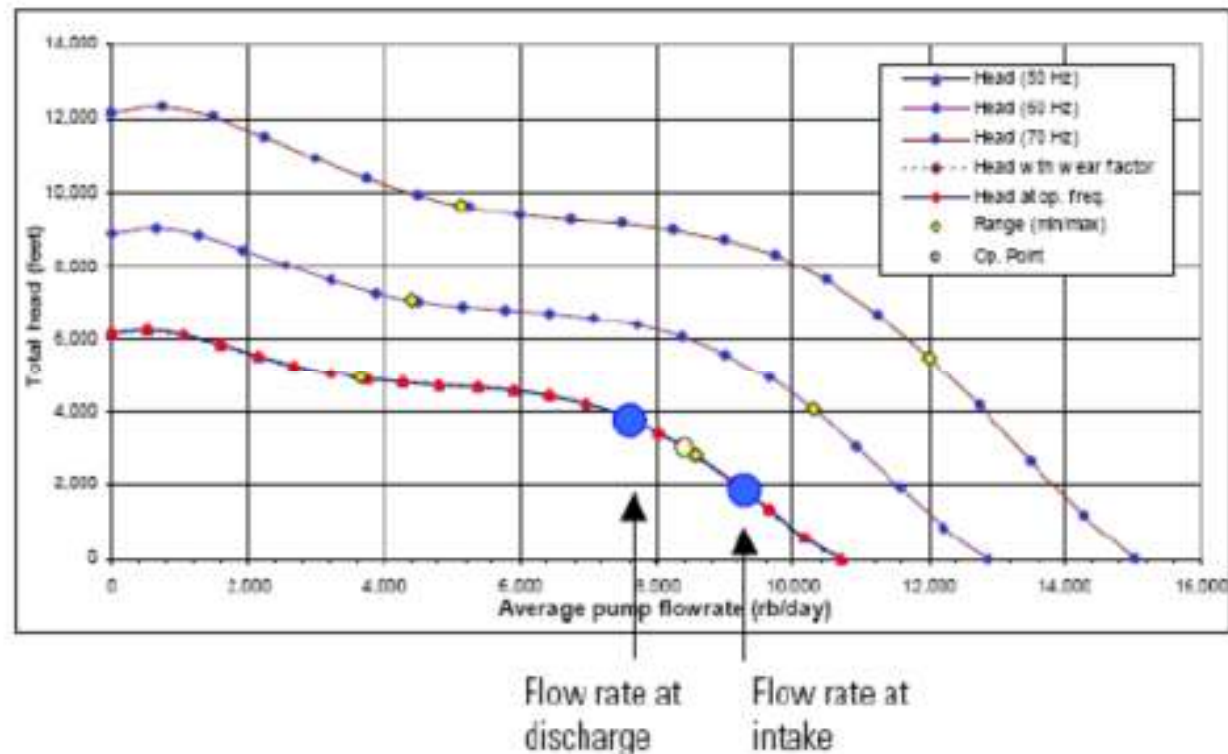
This example shows the effect of a well being produced with an intake pressure below bubble point. It can be seen that the pressure trend oscillates as the volume of gas at the pump intake increases. The oscillation occurs as the pump ΔP decreases due to the lighter density free gas. As the intake pressure rises gas is forced back into solution and the pump delta increases again, producing a lower intake pressure. The total oscillation period is approximately 3 minutes in this case.

Using the MultiSensor data on high GOR wells

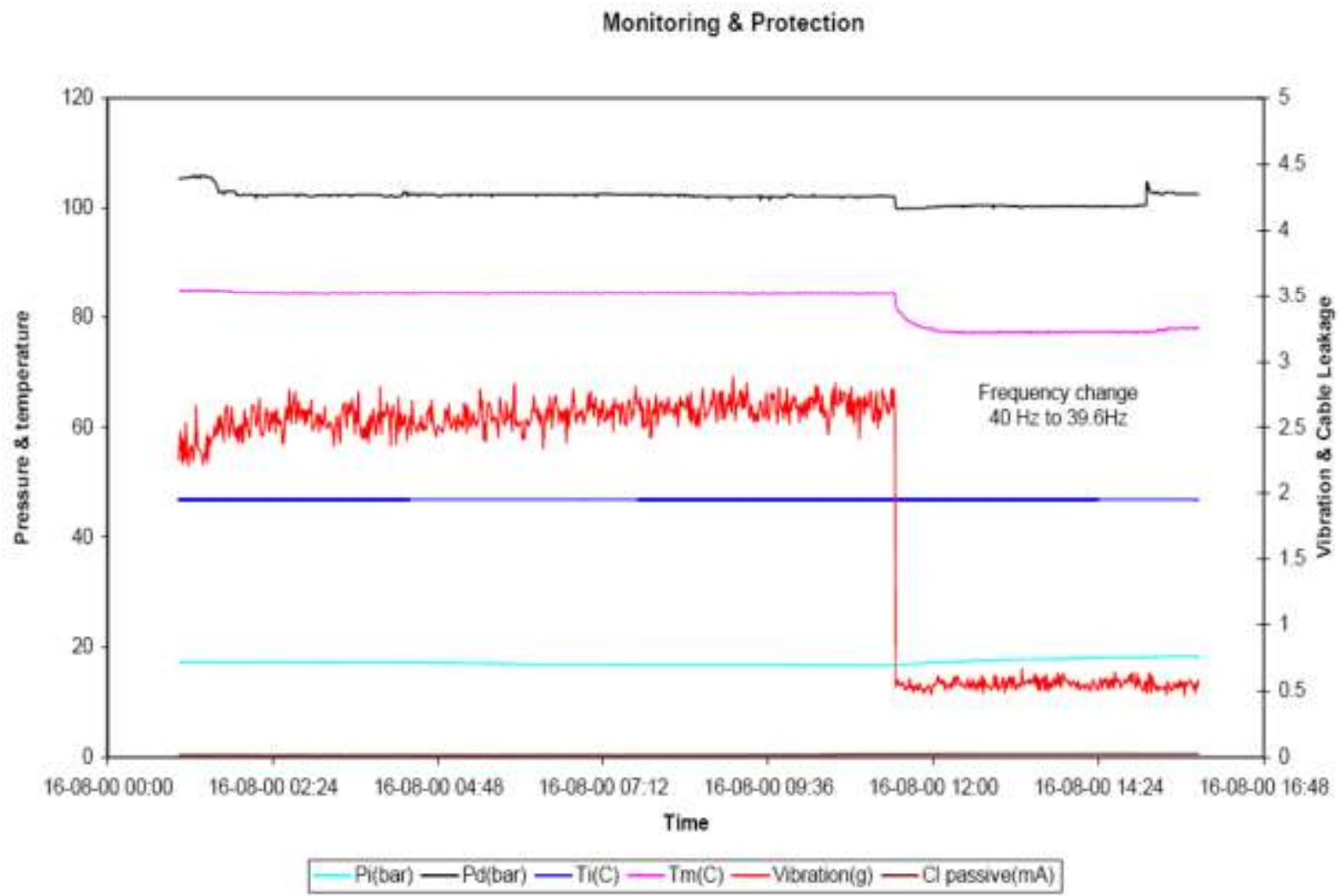


Using the MultiSensor data on high GOR wells

Due to the volume of free gas at the intake the flow rate is higher than that of the pump discharge causing the lower ESP stages to the running in an up thrust condition



Reducing Vibrations



Reducing Vibrations

Trend vibration data has shown that certain ESPCP operating frequencies can produce high vibration readings.

Adjusting the operating frequency by a small amount up or down can significantly reduce the vibration value, reducing the mechanical stress on the downhole completion. The plot also shows the effect on motor temperature as the frequency is reduced. Power consumption of the pump is proportional to the frequency of operation; therefore reducing the speed of the pump reduces the load on the motor.

An identical phenomenon occurs with ESP and PCP pumps which are frequency controlled.

Questions and Answers Session